

Book Management API

Project Overview

Book Management API is a RESTful backend application developed using Laravel 12, MySQL, and JWT Authentication.

This project includes:

- User Registration
 - User Login
 - JWT Authentication
 - Protected Profile API
 - Books CRUD Operations
 - Search Functionality
 - Pagination
 - Soft Delete
 - Form Request Validation
 - Custom JWT Middleware
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Technology Stack

- PHP 8.2+
 - Laravel 12
 - MySQL
 - JWT Authentication (`tymon/jwt-auth`)
 - Postman
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Features

Authentication Module

- User Registration
- User Login
- JWT Token Generation
- Protected Routes using JWT Middleware
- User Profile API

Books Management Module

- Add Book
- View Books List

- Search Books
 - Pagination
 - View Single Book
 - Update Book
 - Soft Delete Book
-

Project Setup

Step 1: Extract Project

Extract the project ZIP file into your desired directory.

Example:

```
C:\xampp\htdocs\company_project
```

Step 2: Install Dependencies

Open terminal inside the project folder and run:

```
composer install
```

Step 3: Environment Configuration

Copy:

```
.env.example
```

to:

```
.env
```

Generate application key:

```
php artisan key:generate
```

Step 4: Database Configuration

Update the database credentials in `.env`:

```
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=company_project
DB_USERNAME=root
DB_PASSWORD=
```

Step 5: Database Setup

Create a database named:

```
company_project
```

Import the provided SQL file:

```
database.sql
```

using phpMyAdmin or MySQL Workbench.

Step 6: JWT Configuration

Generate JWT Secret Key:

```
php artisan jwt:secret
```

Step 7: Run Application

```
php artisan serve
```

Application URL:

```
http://127.0.0.1:8000
```

API Documentation

Base URL

```
http://127.0.0.1:8000/api
```

Authentication APIs

Register User

Endpoint

```
POST /auth/register
```

Request Body

```
{
  "name": "Ashu Raj",
  "email": "ashu@example.com",
  "password": "123456",
  "password_confirmation": "123456"
}
```

Success Response

```
{
  "success": true,
```

```
{
  "message": "User Registered Successfully"
}
```

Login User

Endpoint

```
POST /auth/login
```

Request Body

```
{
  "email": "ashu@example.com",
  "password": "123456"
}
```

Success Response

```
{
  "success": true,
  "token": "JWT_TOKEN"
}
```

User Profile

Endpoint

```
GET /profile
```

Headers

```
Authorization: Bearer JWT_TOKEN
```

Books APIs

Create Book

Endpoint

`POST /books`

Headers

Authorization: Bearer JWT_TOKEN

Request Body

```
{
  "title": "Laravel Basics",
  "author": "Ashuraj",
  "price": 499,
  "published_date": "2025-05-30"
}
```

Get Books List

Endpoint

`GET /books`

Query Parameters

```
search=laravel
limit=10
page=1
```

Example

`GET /books?search=laravel&limit=10&page=1`

Get Single Book

Endpoint

```
GET /books/{id}
```

Example

```
GET /books/1
```

Update Book

Endpoint

```
PUT /books/{id}
```

Example

```
PUT /books/1
```

Request Body

```
{
  "title": "Laravel Advanced",
  "author": "Ashuraj",
  "price": 699,
  "published_date": "2025-05-30"
}
```

Delete Book

Endpoint

```
DELETE /books/{id}
```

Example

```
DELETE /books/1
```

Note

Book records are soft deleted using:

```
_deleted = 1
```

Validation

The application uses Laravel Form Requests for validation.

Validation includes:

- Required fields
- Unique book title validation
- Email format validation
- Unique email validation
- Numeric price validation
- Date validation

Security Features

- Password Hashing using bcrypt
- JWT Authentication
- Protected APIs using Middleware
- Request Validation
- Eloquent ORM
- SQL Injection Protection

Deliverables Included

- Project Source Code
- SQL Dump File
- README.md
- Postman Collection

Author

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